

Hey, everybody. Welcome to your chapter 15 review. We're going to continue to explore each component of integrated training in further detail. Today's topic, cardiorespiratory fitness training.

Cardiorespiratory fitness is vitally important to health and wellness. It reflects the ability of the cardiovascular and respiratory systems to supply oxygen-rich blood to skeletal muscles during sustained physical activity. Cardiorespiratory training should be a top priority when allocating time and resources to client programs because of the number of health-related benefits associated with it, including but not limited to reduced risk of cardiovascular disease, increased longevity, and enhancements to weight management.

Many cardiorespiratory training programs fail to consider rate of progression, which is critical in helping clients achieve their personal health and fitness goals with the most efficient and effective use of time and energy. Rate of progression should be closely monitored, however, to avoid injury or poor adherence if the progression is too fast or too slow respectively.

By the end of this chapter, you should be able to summarize the importance of cardiorespiratory fitness training for different client types with different health and fitness goals, outline the general guidelines for cardiorespiratory fitness training, explain stage training and its use to improve cardiorespiratory endurance, and identify cardiorespiratory training considerations and modifications for clients who have postural conditions.

The benefits of regular physical activity are why we're all here. Research has confirmed an individual's cardiorespiratory fitness level is a strong predictor of morbidity and mortality. There are many, many other great reasons as well, like increased energy, weight loss, and better function.

People can achieve numerous health-related benefits from modest amounts of moderate-intensity exercise and even greater benefits from vigorous-intensity exercise or a combination of both. The current guidelines recommend for adults to perform moderate-intensity exercise, such as brisk walking, at least five times a week, totaling 150 minutes, or accumulate 75 minutes of vigorous-intensity aerobic exercise activity, such as running, at least three days a week. Or it can be an equivalent mix of both intensity levels.

Typically, training sessions include three phases, a warm-up to get the body prepared for physical activity, the conditioning phase, which is your main event, and a cool-down to provide the body with a smooth transition back to a steady state of rest. Just remember that no two people will respond and adapt to cardiorespiratory exercise in exactly the same way, so all training recommendations must be individually determined using the FITTE-VP principal.

FITTE-VP stands for Frequency, Intensity, Type, Time, Enjoyment, Volume, and Progression. Frequency refers to the number of training sessions in a given time period, usually expressed as sessions per week. Intensity refers to the level of demand that a given activity places on the body. Some methods for monitoring cardio exercise intensity include calculating VO2 max using percentages of maximal heart rate or heart-rate reserve, Metabolic Equivalents, or METs, Ratings of Perceived Exertion, a.k.a. RPE, and using the talk test.

Time refers to, as I'm sure you guessed, the length of time engaged in an activity or exercise training session and is typically expressed in minutes. Type refers to the mode of activity, like cycling, running, or swimming, or interval training methods like HITT, Tabata, and Fartlek training. Enjoyment refers to the amount of pleasure derived from engaging in a specific exercise or activity.

Then we get to the VP part. Volume of exercise represents the total amount of work performed in each time frame, typically one week. And finally, progression refers to how an exercise program advances, such as progression in intensity and/or volume.

Cardiorespiratory training programs place different demands on the cardiorespiratory and muscular systems and ultimately affect a client's adaptations and goals. A useful way to break this down further is to study stage training. The purpose of stage training is to ensure cardiorespiratory training programs progress in an organized fashion in order to promote continual adaptation and minimize the risk of overtraining and injury.

Stage training is a five-stage programming system that uses four different training zones, or intensities, to elicit specific physiological adaptations. Stage 1 represents the lowest intensity of exercise and is exclusively within zone 1. It is designed to help improve cardiorespiratory fitness levels using a target intensity below Ventilatory Threshold 1, or VT1, and involves steady-state aerobic exercise.

Use the talk-test method to monitor training intensity. The client should be able to easily hold a conversation during the duration of the activity. It's intended for beginners getting out of a more sedentary lifestyle and is also used as a recovery zone for more advanced clients.

A stage 2 workout consists of a mix of moderate intensity recovery intervals just below VT1 and work intervals performed at an intensity just above VT1. It is performed within zones 1 and 2. Stage 2 is designed for clients with moderate cardiorespiratory fitness levels who are ready and want to begin training at a higher intensity. Once clients become accustomed to stage 2 intervals and have shown positive signs of adapting to the physical demand, they can begin performing moderately intense, steady-state cardio exercise just above VT1 if desired.

A stage 3 workout includes the client moving in and out of training zones 1, 2, and 3, or between light, moderate, and vigorous exercise in a progressive volume. Clients should only progress to this stage after obtaining a satisfactory cardiorespiratory fitness base. It's not appropriate for all clients. Those seeking only modest improvements, those who are new to exercise, and those who are averse to high-intensity training can skip stage 3 training.

Stage 4 introduces zone 4 max-effort exercises and includes performing intervals that involve all four training zones. It's intended for the advanced client who has obtained a high level of cardiorespiratory fitness and seeks improvement in maximal anaerobic performance, such as athletes and top-end fitness enthusiasts. The volume of training should be slow and gradual. And if the client exhibits signs of overtraining, such as disturbed sleep, excessive fatigue, impaired performance, loss of appetite, or mood swings, the volume and intensity of the workouts should be decreased, and recovery time between sessions should be increased.

Stage 5 consists of sport-specific training and is useful for all types of athletes. The focus is on drills to help improve conditioning using linear, multidirectional, and sport-specific activities performed as conditioning and often combining high-intensity interval training with small-sided games and agility drills.

Most weight-loss clients and those new to exercise will stay within stage 1 and 2 for effective health and wellness improvements. These two stages should be a top priority for the average exerciser, whereas stages 3 to 5 are most appropriate for athletes and advanced exercisers who are seeking greater improvements in cardiorespiratory conditioning. It is not necessary to implement all five stages with your clients.

It's also noteworthy to mention that, because movement is involved, it's important to monitor posture during cardio workouts. As people get tired, it's easy to slip into old habits and compensate within the kinetic chain. Some of the common postural deviations you might see include rounded shoulders and forward head, anterior pelvic tilt, and adducted and internally rotated hips, valgus of the knees, and pronated feet. So keep an eye on posture, and make corrections as necessary.

The most important thing is to remember to design the right program for each individual client based on their current abilities and goals and to revisit their progress and goals often so they don't get burned out physically or mentally. You'll also want to hone in on the E in FITTE-VP because enjoyment is how you'll get clients to stick with the program and with you. Remember moderate intensity cardio at least five times a week, vigorous intensity at least three times per week, or a combination of both.

Just get your clients moving, motivated, and on that path to a longer, happier, and healthier life. And with that, I'm sending you off to continue on your path of learning. Stay well, and enjoy your studies.